UNIT-1: Calculus (Multivariable Calculus & Applications)
Appearance Year: 2025-26

1. Partial Derivatives:
   - If degree of u(x,y) = (sqrt(x)+sqrt(y))^5, find x(du/dx) + y(du/dy).
   - If z = x^4 y^2 sin^-1(x/y) + log x - log y, show x(dz/dx) + y(dz/dy) = 6 x^4 y^2 sin^-1(x/y).
2. Jacobians:
   - If x = r cos(theta), y = r sin(theta), find Jacobian J(r,theta)/(x,y).
3. Taylor's Series:
   - Expand x^2 y + 3y - 2 in powers of (x-1) and (y+2) up to 3rd degree.
4. Maxima & Minima / Lagrange Multipliers:
   - Find minimum distance from (1,2,0) to cone z^2 = x^2 + y^2.
5. Approximation of Errors:
   - Find approximate value of [(0.98)^2 + (2.01)^2 + (1.94)^2]^(1/2).